

# SOFTWARE ENGINEERING MINI-PROJECT

## SYNOPSIS / PROJECT PROPOSAL

### 1. Project Title and Group Details

**Project Title:** Smart Complaint Management System for a community

**Group Number:** Team 9

**Course:** UE24CS341A – Software Engineering

**Development Methodology:** Agile

**Customer / Stakeholder:** Resident, Administrator, Maintenance staff / Association Management (mechanic, plumber, etc)

### 2. Team Profile and Product Ownership

| Team Member            | SRN           | Proposed Owned Functionality               | Secondary Responsibility      |
|------------------------|---------------|--------------------------------------------|-------------------------------|
| Riya Saju Vithayathil  | PES1UG24CS378 | User Management and Complaint Registration | Requirements Engineering      |
| Ritu Ranganath Kashyap | PES1UG24CS377 | Complaint processing and resolution        | Testing and quality assurance |
| Rohan Suresh           | PES1UG24CS383 | Administration, Feedback and Reporting     | Documentation & CI/CD         |
| Rohit Krishna R        | PES1UG24CS385 | Complaint assignment and tracking.         | System design & UML           |

**Note:** The above division is preliminary and may be refined during sprint planning.

Each member will participate in all major Software Development Life Cycle activities while taking primary ownership of at least one complete functional area. This follows the project requirement that every member contribute to implementation and own complete functionality rather than considering UI, database interaction or testing alone as an individual functional feature.

### 3. Proposed Project Description

The **Smart Complaint Management System** is a web-based application designed to provide a centralized and systematic platform for registering, assigning, tracking and resolving complaints within an organization or institution.

In conventional complaint-management processes, complaints may be communicated through emails, phone calls, paper forms or other informal channels. This can make it difficult for users to know whether their complaint has been acknowledged, who is responsible for handling it and what progress has been made. Similarly, administrators and staff may face difficulties in organizing complaints, assigning responsibilities and monitoring unresolved issues.

The proposed system will digitize the complete complaint-management process. Registered users will be able to submit complaints by providing relevant information such as the complaint title, category, description, location and supporting attachments. Each successfully submitted complaint will be assigned a **unique Complaint ID**, which can be used to track its progress.

Administrators will be able to review complaints and assign them to the appropriate department or staff member. Authorized staff will be able to view complaints assigned to them, update their status, add remarks and record the final resolution. Users will be able to monitor these status changes and provide feedback after their complaint has been resolved.

The system will therefore provide a structured workflow from **complaint registration to final resolution and closure**, improving transparency, accountability and communication among users, staff and administrators.

## 4. Problem Statement

Existing complaint-handling methods can become inefficient when complaints are received through multiple disconnected channels. Users may have limited visibility regarding the progress of their complaints, while staff may find it difficult to organize, assign and monitor a large number of complaints.

The absence of a centralized system can result in delayed responses, unclear responsibility, loss of complaint information and difficulty in evaluating how effectively complaints are being resolved.

The Smart Complaint Management System aims to address these problems by providing a single platform through which complaints can be **submitted, assigned, processed, tracked, resolved and reviewed**.

## 5. Intended Users

The system will primarily support the following user classes:

### User / Complainant

A user can register and log in to the system, submit complaints, upload supporting information, view previously submitted complaints, track complaint status and provide feedback after resolution.

## **Staff / Officer**

Authorized staff members can view complaints assigned to them, examine complaint details, update the processing status, add remarks and record resolution details.

## **Administrator**

The administrator manages the overall complaint-management process. The administrator can manage users, complaint categories and departments, assign complaints to responsible staff, monitor complaint progress and view reports and statistics.

## **6. Objectives**

The major objectives of the Smart Complaint Management System are:

- To provide a centralized platform for registering and managing complaints.
- To provide every complaint with a unique identifier for easy tracking.
- To allow users to monitor the current status and progress of their complaints.
- To enable complaints to be assigned to the appropriate department or staff member.
- To enable authorized staff to systematically process and resolve assigned complaints.
- To maintain a history of complaint assignments and status changes for accountability.
- To notify users about important changes in the complaint lifecycle.
- To enable administrators to monitor pending and resolved complaints.
- To collect user feedback regarding complaint resolution.
- To develop the system incrementally using Agile software development practices.

## **7. Functional Features**

At the completion of the project, the proposed system is expected to provide the following major functional features:

1. **User Registration and Authentication**  
Users can create accounts, securely log in and access functionality according to their assigned role.
2. **Complaint Registration**  
Authenticated users can submit complaints containing a title, description, category, location and optional supporting attachments.
3. **Unique Complaint Identification**  
Every successfully submitted complaint is assigned a unique Complaint ID.
4. **Complaint Categorization**  
Complaints can be organized into predefined categories to assist in management and assignment.
5. **Complaint Assignment**  
Administrators can assign complaints to the appropriate department or responsible staff member.

6. **Complaint Processing and Status Update**

Assigned staff can process complaints, add remarks and update their status through stages such as **Submitted, Assigned, In Progress, Resolved and Closed**.

7. **Complaint Tracking**

Users can view the current status and relevant history of complaints submitted by them.

8. **Search and Filtering**

Authorized users can search and filter complaints based on criteria such as Complaint ID, category, status, department and date.

9. **Notifications**

Users can receive notifications when significant events occur, such as complaint submission, assignment, status changes and resolution.

10. **Complaint Resolution and Closure**

Staff can record resolution details, and resolved complaints can subsequently be closed according to the defined workflow.

11. **Feedback and Rating**

Users can provide feedback or a rating after their complaint has been resolved.

12. **Administrative Dashboard and Reporting**

Administrators can view summary information regarding total, pending, in-progress and resolved complaints and generate basic reports.

## 8. Scope,

### In Scope

- User registration and authentication
- Role-based access for users, staff and administrators
- Complaint registration
- Complaint categorization
- Complaint assignment
- Complaint status management
- Complaint tracking and history
- Search and filtering
- Supporting file attachments
- Notifications
- Resolution recording
- User feedback and ratings
- Administrative dashboard
- Basic complaint reports

### Out of Scope

The following features are not part of the initial implementation and may be considered as future enhancements:

- AI-based automatic complaint classification
- AI-based priority prediction

- Automatic duplicate complaint detection
- SMS/WhatsApp integration
- Native Android/iOS applications
- Advanced geographical complaint mapping
- Advanced predictive analytics
- Large-scale multi-organization deployment

## 9. Short-Term Plan of Work

The project will be developed incrementally using **Agile methodology**. The preliminary sprint plan is as follows:

| Sprint          | Planned Work                                                                       | Expected Evidence                                                     |
|-----------------|------------------------------------------------------------------------------------|-----------------------------------------------------------------------|
| <b>Sprint 1</b> | Project setup, requirement analysis, user stories, GitHub setup and authentication | Requirements baseline, backlog, repository and working authentication |
| <b>Sprint 2</b> | Complaint registration, complaint categories and complaint viewing                 | Working complaint submission module and unit tests                    |
| <b>Sprint 3</b> | Complaint assignment and staff complaint-processing workflow                       | Functional assignment and status-update workflow, PRs and tests       |
| <b>Sprint 4</b> | Complaint tracking, notifications, resolution and feedback                         | Integrated complaint lifecycle and test evidence                      |
| <b>Sprint 5</b> | Admin dashboard, search/filtering, reporting and security improvements             | Working admin features, security validation and coverage report       |
| <b>Sprint 6</b> | Integration testing, system validation, bug fixing and final deployment            | Final working system, validation evidence and project report          |

The exemplar specifically expects the short-term plan to connect each iteration with a **planned output and acceptance evidence**, rather than merely listing dates.

## 10. Proposed Technology and Tool Stack

| Activity        | Proposed Technology / Tool |
|-----------------|----------------------------|
| Backend         | Python – Flask             |
| Frontend        | HTML, CSS, JavaScript      |
| Database        | MySQL                      |
| Version Control | Git                        |

|                          |                              |
|--------------------------|------------------------------|
| Repository               | GitHub                       |
| Agile Project Management | Jira / GitHub Projects       |
| CI/CD                    | Jenkins                      |
| Containerization         | Docker                       |
| Testing                  | pytest                       |
| API Testing              | Postman                      |
| Static Code Analysis     | SonarQube or similar         |
| UML / Design             | Draw.io / PlantUML / StarUML |

These tools align with the course infrastructure, which explicitly identifies Git/GitHub, Jenkins, Docker, testing tools, Jira or similar tools, and SonarQube or similar static-analysis tooling.

## 11. Agile Methodology

The project will follow an **Agile development methodology**, as required for the Software Engineering mini-project.

Development will be divided into multiple iterations or sprints. Requirements will be represented as user stories and maintained in the product backlog. At the beginning of each sprint, selected user stories will be divided into development tasks and assigned to team members.

GitHub will be used for version control and collaborative development, while Jira project-management tool will be used to maintain the product backlog, sprint tasks and progress. Features will be implemented, reviewed and tested incrementally rather than developing the complete application in a single phase.

This approach will allow the team to continuously evaluate progress, identify defects early, accommodate requirement refinements and maintain traceability between requirements, implementation and testing.

## 12. Expected Outcome

At the end of the project, the team expects to deliver a functional web-based **Smart Complaint Management System** that supports the complete basic complaint lifecycle:

**Complaint Submission → Assignment → Processing → Resolution → Feedback → Closure**

The completed system should provide users with greater visibility into complaint progress, provide staff with an organized mechanism for handling assigned complaints and provide administrators with centralized control and monitoring.

In addition to the working software, the project will demonstrate the major stages of the Software Development Life Cycle through requirements specification, Agile planning, architecture and design, implementation, testing, security validation, CI/CD and project documentation, which are all emphasized in the course guidelines.